STAMPED properties for reproducible research objects

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1 Abstract

2 Introduction

Data-driven research depends on specific versions of datasets, software, and computational environments, together with a record of how they were combined to produce results. We adopt the term **research object** to denote a collection of data, code, and metadata that together represent the research output as a unit. It is common for the components of a research object to be managed separately—code in one repository, data on a shared drive, environment setup in a wiki page, provenance in a lab notebook or nowhere at all—and this fragmentation undermines rigor, reproducibility, reusability, and efficiency. Even when gathered in one place, research objects are often tightly coupled to specific environments, opaque to outside collaborators, and difficult to reuse or redistribute.

The FAIR principles[1] and FAIR4RS guidelines[2] provide structured guidance for making digital objects findable, accessible, interoperable, and reusable. The Workflow Community Initiative (WCI-FW)[3] mapped these principles onto computational workflows but explicitly chose not to define new ones, instead treating workflows as hybrid data/software objects. FAIR addresses how digital objects should be described so they can be found, accessed, and reused. STAMPED addresses a complementary question: how a research object should be organized and managed so it can actually be re-executed, extended, and redistributed. Both frameworks cover overlapping concerns—for example, persistent identifiers serve both Findability and Tracking—but accent different aspects: FAIR emphasizes discovery and interoperability across repositories, while STAMPED emphasizes improving the operational maturity of scientific workflows in day-to-day research practice.

Many researchers already employ practices that address these operational properties—version control, containerized environments, structured project directories, documented analysis steps—but lack a shared vocabulary to name them, evaluate how thoroughly they are applied, and communicate that evaluation to collaborators and reviewers. Here we provide that vocabulary: the STAMPED properties (Self-contained, Tracked, Actionable, Modular, Portable, Ephemeral, and Distributable), originating from the YODA Principles[4], which describe the operational maturity of a research object. Rather than a compliance threshold, each STAMPED property is a spectrum from a practical minimum—often what researchers are already doing—to an aspirational ideal. A researcher who stores everything under one project directory, version-controls analysis scripts, and documents how to run them is already meeting the practical minimum for Self-containment, Tracking, and Actionability. As complexity grows—multi-institutional collaborations, heterogeneous workflows spanning platforms, long-term reproducibility requirements—the same properties guide researchers toward more rigorous practices without requiring wholesale adoption of new tools. STAMPED provides a vocabulary for describing and a framework for evaluating and incrementally improving the operational properties of research objects toward enhanced rigor, reproducibility, reusability, and efficiency.

3 Comments

The items containing keywords "MUST", "SHALL", "SHOULD", and "MAY" are to be interpreted as described in RFC 2119[5].

Table 1: The STAMPED Properties. Each property addresses a distinct dimension of research object organization that contributes to reproducibility. Together, they provide a framework for evaluating and improving the reproducibility of computational research.

Letter	Principle	Description
S	Self-containment	A research object is a complete retrieval unit—it can be obtained and understood in its intended scope without needing to reference external resources.
T	Tracking	The state and provenance of all components is recorded.
A	Actionability	Research object contains machine-actionable information to carry out procedures to obtain or reproduce content.
M	Modularity	All modules are independent and composable.
P	Portability	Research object could be moved to different environments while retaining its STAMPED properties.
E	Ephemerality	Procedural execution is performed within a throwaway environment.
D	Distributability	All modules and procedures are shareable externally in a persistent state.

3.1 Self-containment

- **S.1**: All modules and components essential to replicate computational execution MUST be reachable within a single top-level research object.

Reproducibility fails when a research object depends on resources that are not part of it—an undocumented library, a dataset referenced by a broken URL, a script that assumes tools are installed on the host system. We call this the "don't look up" rule: a research object must never rely on implicit external state. Instead, it must be a complete retrieval unit, where all modules and components essential to reproduce its results are contained within a single top-level boundary (S.1, Table 1).

Components may be included literally (e.g., files committed directly) or by reference (e.g., as subdatasets, registered data URLs). Both approaches satisfy self-containment, provided the references are explicit and part of the research object. At a minimum, everything needed is gathered under one root and any external dependency is explicitly documented—ensuring that nothing is implicitly assumed about the host system, a concern further addressed by Portability. At the ideal end, every reference is precise enough that retrieval is unambiguous; how to achieve that precision is the concern of Tracking (T) and Distributability (D).

Self-containment is the foundational property upon which the remaining STAMPED properties build. Tracking, Modularity, Portability, and Distributability each address a different aspect of how that boundary is organized, versioned, and shared—but none are meaningful without first establishing what is inside it.

3.2 Tracking

- **T.1**: Persistent content identification MUST be recorded for all components.
- **T.2**: All components SHOULD be tracked using the same content-addressed version control system.
- **T.3**: Provenance of all modifications MUST be recorded.
- **T.4**: Code-driven provenance SHOULD be recorded programmatically and MUST include the versions of all components involved.

Research data and code change constantly during a project, and each change compounds the difficulty of reconstructing any previous state: which version of a dataset was used for a particular analysis, what parameters were set, what code produced a given figure. Tracking addresses two concerns: the identity of each component—which exact state it was in—and **provenance**, the recorded history of what actions produced or modified it. These are often managed separately—versions in one system, provenance in a lab notebook or nowhere at all—and this fragmentation makes prior work hard to validate.

The spectrum of Tracking begins with content-addressed version control. Rather than relying on semantic version labels, content-addressed systems identify each state by a checksum of its contents. This distinction matters: two datasets labeled "version 1.0" by different labs are ambiguous; two datasets with identical content hashes are provably identical. Version control commits also capture basic provenance—who made the change, when, and a description of why—unifying identity and provenance in a single system. This also provides a safety net: any change can be reverted and any prior state restored, making it safe to modify and experiment. For tracking to be effective, all components of the research object must be under version control—code, data, environment definitions, and configuration.

Traditional version control is designed for code and other text files, but research objects often include large binary datasets, container images, and other assets that do not fit this model. Compatible tools such as git-annex, DataLad, or DVC extend content-addressed tracking to these components, ensuring that all parts of the research object are managed under the same system.

Toward the ideal end, provenance is captured programmatically by tooling rather than by manual annotation, producing richer records than a human would typically write: each computational step records what command was executed and what inputs it consumed. When modules are independently tracked under compatible systems (M), the version of every module at the time of each commit is also discoverable, providing a complete picture of the state that produced a given result. This enables not just inspection but re-execution.

3.3 Actionability

- **A.1**: Research object MUST contain sufficient instructions to reproduce all computational results.
- **A.2**: Procedures SHOULD be specified as executable specifications.

A research object may be self-contained and fully tracked, yet having all ingredients and a record of what was done is insufficient without a recipe that can be followed. Requirement A.1 is what makes a research object operationally useful—the bridge between having the components and being able to use them.

At a minimum, a research object must contain documentation thorough enough for another researcher to follow every step to reproduce results. Actionability applies to every other STAMPED dimension: at the ideal end, each property is enacted not through documentation alone but through executable specifications (A.2):

- **Self-containment** is more actionable when retrieval of all components is specified and executable (e.g., `git clone`, `datalad install`), not just listed in a manifest.
- **Tracking** is more actionable when provenance records are re-executable (e.g., `datalad rerun`), not just inspectable.
- **Modularity** is more actionable when components can be composed, updated, and replaced via tooling (e.g., `git submodule`), not just organized into directories.
- **Portability** is more actionable when environment specifications are machine-readable and can be resolved on different hosts (e.g., `conda env create`, Nix flakes), not just documented in a README.
- **Ephemerality** is more actionable when computation can be orchestrated in disposable environments (e.g., `docker compose`, Slurm), not just instructed to "run in a clean environment."
- **Distributability** is more actionable when a frozen state can be produced and retrieved by others (e.g., containers, `npm ci`), not just described with pinning instructions.

The principle is tool-agnostic: any system that moves a property from documented to executable moves the research object further along the actionability spectrum.

3.4 Modularity

- **M.1**: Components SHOULD be organized in a modular structure.
- **M.2**: Components MAY be included directly or linked as subdatasets.

Separation of concerns is a foundational engineering practice: organizing a system so that each piece has a clear, distinct role. With clear boundaries between the conceptual components of a research object, it becomes more navigable and maintainable. We define a **module** as a separately distributable unit of a research object—such as a dataset, a collection of analysis scripts, or a computational environment definition. In practice, module boundaries often align with repository boundaries, but a module may also be a published container, a registered dataset, or any independently versioned unit. Modularity applies this principle to research objects, structuring them as assemblies of modules whose boundaries are explicit. At a minimum, a research object should separate analysis code, input datasets, computational environments, configuration settings, and results into distinct directories. Beyond this, independent versioning of modules enables reuse: a dataset or environment can be updated to a newer version, or swapped for an alternative, without disrupting the rest of the research object. This is possible through manual management but becomes practical when handled by tooling that automates module installation, versioning, and composition. Toward the ideal end, the full research object can be reassembled from its specification through automated, recursive installation of its modules.

3.5 Portability

- **P.1**: Procedures MUST NOT depend on undocumented host environment state.
- **P.2**: Computational environments MUST be explicitly specified.
- **P.3**: Environment definitions MUST be version controlled.

A research object that is self-contained, tracked, and modular may still fail to run on a different machine if it is silently affected by inherited host state—a specific Python installation, a system library, a hardcoded path, or an assumed OS configuration. This coupling is often invisible on the original system and only surfaces when someone else attempts to reproduce the work. Portability requires making all host dependencies explicit, so that the research object can be adapted to a new environment.

Some degree of coupling to the host is unavoidable—path naming restrictions, resource limits, platform-specific flags. The spectrum of Portability begins with isolating this host-specific configuration from the rest of the research object, so that adapting to a new environment, however involved that may be, requires only changing configuration. Beyond this, several complementary approaches reduce host coupling: virtual machines provide full system-level isolation; container-based tools (Docker, Singularity/Apptainer) bundle OS-level environments; and declarative package managers (Nix, Guix) specify environments that can be reconstructed from a definition. Toward the ideal end, the computational environment is fully specified and version controlled within the research object, enabling it to be instantiated on any compatible system.

3.6 Ephemerality

- **E.1**: Computational results SHOULD be produced in ephemeral environments.

A research object may satisfy Self-containment, Tracking, and Portability while its computational environment has only ever been assembled once, incrementally, on the original researcher's machine. During development, researchers routinely modify their environment—installing packages, adjusting configurations, setting variables—and each change is individually small but collectively difficult to document. The result is environment drift: an environment that works but may not be reconstructable from its specification alone, where undocumented steps can go unnoticed when internalized knowledge silently fills the gaps. Ephemerality eliminates this problem by construction: a disposable environment built from tracked specifications cannot carry undocumented state, so any successful execution exercises Self-containment, Actionability, and specification completeness.

The spectrum of Ephemerality ranges from manually setting up a fresh environment from the project's specification—sufficient to catch environment drift—to fully automated disposable environments created and destroyed per execution. At the ideal end, ephemeral computation also enables testing across different host systems, where same-host execution validates specification completeness and different-host execution validates that specifications are not coupled to a specific system. The FAIRly big workflow approach[6] and platforms

such as Code Ocean operationalize this ideal, treating each computation as an ephemeral work unit—which also enables scaling, as independent disposable instances can be parallelized across subjects, parameters, or datasets. Ephemeral computation does not validate everything—for instance, an ephemeral environment with network access may fail to notice unpinned dependencies fetched at build time, a self-containment gap that only surfaces when those resources move or disappear—but it substantially raises confidence that the research object is self-contained, portable, and actionable.

3.7 Distributability

- **D.1**: All referenced modules and components MUST be persistently retrievable by others.
- **D.2**: Environment specifications SHOULD support reproducible builds.

A research object that is self-contained on the author's machine but cannot be obtained by others in the same state is effectively unreproducible beyond its origin. Self-containment (S) establishes that everything needed is within the boundary and Tracking (T) records the state of each component. Distributability ensures that others can obtain the research object in that same state.

The distinction mirrors the concept of a software distribution: a curated, versioned bundle in which all components are resolved to specific versions and packaged for consumption. Simply sharing a research object—uploading scripts to a repository with loose dependencies, or posting files on a website with no version guarantees—does not constitute distribution in this sense. A distributable research object is packaged so that it can be retrieved in the same state as intended.

Distributability also has a circular relationship with the other STAMPED properties: someone else's distribution effort often serves as the starting point for a new research object's self-containment. Researchers routinely begin by downloading containers, fetching published datasets, and installing released software—modularly composing the distributions of others to assemble their own self-contained research objects.

The spectrum of Distributability begins where FAIR leaves off: publicly accessible components with documented retrieval instructions. Toward the ideal end, several complementary strategies reduce the risk that external changes will break a research object's reproducibility: bundling dependencies into containers or archives reduces the surface area of components that can independently change or disappear; hosting on archival infrastructure (Zenodo, DANDI, Software Heritage) with content-addressed identifiers allows recipients to verify integrity regardless of source; and mirroring across multiple platforms protects against the failure of any single registry.

3.8 Checklist for compliance to principles

We hope that the preceding sections have provided a clear understanding of the STAMPED properties and their rationale. While this understanding is essential, researchers may also benefit from a practical checklist to assess their research objects against these principles. This checklist provides such a guide for assessing compliance with STAMPED principles, ordered by the strength of the requirement (MUST, SHOULD, MAY).

MUST

- ▷ **Self-containment (S.1)**: All modules and components essential to replicate computational execution MUST be reachable within a single top-level research object.
  - ☐ Are all files and directories nested under a common root?
  - ☐ Are datasets included, linked, or referenced reachable from the code and environment specifications without cross the boundary of a common root?
  - ☐ Is external software included in the environment or linked as submodules within the project boundary?
- ▷ **Tracking (T.1)+(T.3)**: Persistent content identification MUST be recorded for all components. Provenance of all modifications MUST be recorded.

Table 2: Normative statements about STAMPED properties of research objects.

Principle	Requirements
To be Self-contained:	• all modules and components needed to understand and execute the Research Object are retrievable as a single unit. • external dependencies are explicitly documented with retrieval instructions. • there are no implicit references to undocumented external resources.
To be Tracked:	• every component has version information (commit hash, tag, or identifier). • changes to components are recorded with timestamps and authorship. • provenance records capture the computational history, context, and transformations.
To be Actionable:	• instructions for executing procedures are present and unambiguous. • execution paths can be followed manually or automated programmatically. • the Research Object transitions from documentation to operational capability.
To be Modular:	• modules can be independently modified. • components are organized in logical, separable units. • modules can be composed together or used in isolation.
To be Portable:	• system requirements and dependencies are explicitly documented. • the Research Object is flexible enough to execute on different host environments without modification by the user. • environment specifications are machine-readable where possible.
To be Ephemeral:	• computation can occur in temporary, disposable environments. • results are reproducible without knowledge of previous runs. • no reliance on external configurations or host system states (such as OS registry modifications).
To be Distributable:	• all modules and components can be shared in a persistent, retrievable state. • dependencies are frozen or pinned to specific versions across systems. • the Research Object can be obtained by others in the same state as intended.

- ☐ Are version control systems such as `Git` used for code, text, documentation, and configuration files?
- ☐ Are version control systems such as `git-annex`, DataLad, or `Git` LFS used for large binary data?
- ☐ Are the exact environment specifications used to generate a set of results included in the provenance records to link computational actions to a particular environments?

▷ **Actionability (A.1)**: The research object MUST contain sufficient instructions to reproduce all computational results.

- ☐ Is a `README.md` or `Makefile` included with instructions for installation and usage?
- ☐ Is there a clear starting point for users to start reproducing results (e.g., a main script, a workflow definition, or a container image)?

▷ **Portability (P.1)**: Procedures MUST NOT depend on undocumented host environment state.

- ☐ Are relative paths used in scripts, avoiding hardcoded or system-specific paths like `C:\Users\...` or `/home/user/...`?
- ☐ Are all software dependencies included in the environment specification, rather than relying on pre-installed tools?
- ☐ Have all assumptions about the host system's configuration been documented (e.g., specific OS versions, required system libraries, or environment variables)?

▷ **Portability (P.2)**: Computational environments MUST be explicitly specified.

- ☐ Is there a clear list of system requirements and dependencies documented in the README or environment specifications?

▷ **Portability (P.3)**: Environment definitions MUST be version controlled.

- ☐ Are environment specifications (e.g., Dockerfiles, `pyproject.toml`, `package.json`) included in the version control system alongside code and data?
- ☐ Is there a process for updating environment specifications when dependencies change, and are these updates tracked in version control?

▷ **Distributability (D.1)**: All referenced modules and components MUST be persistently retrievable by others.

- ☐ Are environment specifications (e.g., container digests, frozen package manifests) included to ensure others can attempt to exactly replicate the environment?
- ☐ Are environment specifications shared in a way that others can access and use them (e.g., published container images, archived environment files)?
- ☐ Is there documentation on how to obtain and use the environment specifications for reproduction?

SHOULD

▷ **Tracking (T.2)**: All components SHOULD be tracked using the same content-addressed version control system.

- ☐ Is a common version control system (e.g., `Git`, DVC, `git-annex`, or DataLad) used across all components?

▷ **Actionability (A.2)**: Procedures SHOULD be specified as executable specifications.

- ☐ Is the workflow tested regularly to ensure instructions remain accurate?

▷ **Modularity (M.1)**: Components SHOULD be organized in a modular structure.

- ☐ Are raw data, processed data, code, and environment definitions separated into distinct modules?

▷ **Ephemerality (E.1)**: Computational results SHOULD be produced in ephemeral environments.

- ☐ Is the pipeline tested in a fresh container, batch job, clean virtual machine, or cloud-based instance?
- ☐ Does the workflow spawn disposable environments for each execution, allowing for isolation of runs?

▷ **Distributability (D.2)**: Environment specifications SHOULD support reproducible builds.

- ☐ Are the exact environment specifications used to generate a specific set of results regularly tested to ensure they can be reconstituted as intended?
- ☐ Are environment artifacts archived within the research object where possible (e.g., executable binaries, container images)?
- ☐ Is there a process for updating environment specifications when dependencies change, and are these updates tracked in version control?

MAY

▷ **Modularity (M.2)**: Components MAY be included directly or linked as subdatasets.

- ☐ Are external datasets or software dependencies included as submodules or linked as submodules?
- ☐ Is the modular structure documented to clarify how components relate and can be recombined?
- ☐ Are modular boundaries defined in a way that allows independent updates without breaking the overall workflow?
- ☐ Is there a clear mechanism for composing modules together (e.g., `git` submodules, or container orchestration)?
- ☐ Is there documentation or tooling to support users in understanding and navigating the modular structure?

3.9 Enabling tools

No single tool addresses all the dimensions of scientific rigor described above, though many tools contribute to one or several properties simultaneously. For example, container technologies such as Docker or Singularity serve Portability by providing OS-level isolation from explicit environment specifications, enable Ephemerality by disposing per-execution environments, and facilitate Distributability by producing frozen, shareable artifacts. Similarly, version control systems such as `Git` underpin both Tracking (content-addressed identification and change history) and Self-containment (gathering all components under a single root). In practice, self-containing all data might be undesired or prohibitive due to size or other concerns. Extensions such as `git-annex`, `DataLad`, `DVC`, and `Git LFS` strike the balance between the two properties by tracking metadata that identifies specific data versions and obtainability information instead, e.g., content checksums and URLs of remote servers that act as a source, and providing actionable interfaces to obtain the data when necessary.

This overlap is not incidental — it reflects the interdependence of the STAMPED properties themselves. Achieving Ephemerality, for instance, inherently exercises Portability and Actionability: a disposable environment must be reconstituted from explicit specifications, encouraging end-to-end automation. Likewise, Distributability builds on Self-containment and Tracking: a research object can only be shared in a consistent, retrievable state if its boundary is well-defined and all components are precisely identified. The tools that serve multiple properties do so precisely because they operationalize these connections.

Table 3 maps some commonly used tools and technologies to each STAMPED property. A comprehensive up-to-date review of existing tools is out of scope for this formalization paper and would quickly become outdated. To fill that niche, we initiated a satellite project with a website to describe tools and examples, and characterize them in terms of STAMPED and FAIR principles [cite https://myyoda.github.io/principles-examples/stamped_principles/].

STAMPED properties are tool-agnostic. For example, any system that moves a property from documented to executable moves the research object further along the actionability spectrum, but certain tool categories recur across multiple properties; a well-chosen tool can address several dimensions at once, and a thoughtfully assembled toolchain can cover all of them. Researchers should select tools based on their domain, infrastructure, and team familiarity, recognizing that the tools listed here are illustrative rather than prescriptive: what matters is that the underlying property is satisfied, regardless of which tool achieves it.

Where multiple tools serve the same STAMPED property, the choice between them often involves trade-offs between flexibility and simplicity. A representative example is the choice between `git-annex` and Git LFS for large-file tracking. Both appear under Self-containment and Tracking in Table 3, but they occupy different points on a complexity-flexibility spectrum. `git-annex` supports many remote types (e.g. other git repositories over SSH or HTTP, or simple cloud data stores such as S3, local drives, offline USB archives), making it well-suited to multi-institutional collaboration, long-term archival where data sovereignty or offline access is required, as well as local content tracking. Git LFS offers a simpler alternative: it is transparent to users, natively supported by GitHub and GitLab, and broadly adopted. But it is centralized (requiring an LFS server), lacks offline capability, and provides less flexible remote configurations. Lightweight STAMPED implementations using Git LFS are entirely feasible for easier onboarding, trading federation flexibility for reduced operational overhead. The same pattern—flexibility versus simplicity—recurs across other tool choices in Table 3: container-based versus package-based environment management, workflow engines versus shell scripts, and federated archives versus centralized platforms.

Table 3: Sample tools to improve scientific practice for each STAMPED property.

Property	**Tools / Technologies**	**Role**
S — Self-containment	`Git` submodules, DataLad, `git-annex`, DVC, Git LFS	Gather all components (code, data, environments) under a single root via direct inclusion or explicit references (e.g., subdatasets, registered URLs)
T — Tracking	`Git`, `git-annex`, DataLad, DVC, Docker and other containers, Kedro, Git LFS, OSF.io, Zenodo and other data archives providing DOIs	Version control for code and data; content-addressed identification; provenance recording (e.g., `datalad run` for programmatic provenance capture); persistent identifiers for content (DOIs)
A — Actionability	Make, Snakemake, Nextflow, CWL, `datalad run`, `git annex addcomputed`	Define executable specifications for workflows; provide clear entry points for reproducing results
M — Modularity	`Git` submodules, DataLad subdatasets, Kedro	Organize components into independently versioned, composable modules; provide project templates with logical directory separation
P — Portability	Docker, Singularity, Nix, conda, `pip` (`pyproject.toml`, `requirements.txt`), npm	Explicitly specify and isolate computational environments; container-based (OS-level isolation) or package-based (declarative, reproducible builds)
E — Ephemerality	Docker, Singularity, `docker compose`, Slurm, cloud deployments (e.g., AWS, GCP, Azure, GitHub Actions)	Spawn disposable, temporary environments per execution; validate that specifications are complete by reconstituting from scratch
D — Distributability	Zenodo, PyPI, conda-forge, DockerHub/GHCR, RO-Crate, `npm ci`, `pip freeze`, executable compilers	Archive and persistently host frozen, versioned research objects and their components; enable retrieval by others in the intended state

3.10 Examples and Case Studies

Several research projects have already adopted and documented YODA principles—the predecessor to STAMPED—demonstrating practical utility across domains.

BIDS Standard Evolution: A significant validation of the "do not look up" principle occurred within the Brain Imaging Data Structure (BIDS)[7] community. Originally, the BIDS specification nested derived data under `derivatives/` within the original raw dataset, creating upward dependencies that violated portability. Following YODA principles, the BIDS community reversed this relationship: derivative datasets now exist independently and reference raw data as subdatasets, not vice versa.

This architectural shift influenced major neuroimaging tools:

- **fMRIPrep**: Switched default output layout to follow YODA structure
- **OpenNeuroDerivatives**: Entire derivative dataset collection now follows YODA organization, separating processed outputs from raw data dependencies
- **BIDS specification**: Updated to accommodate and recommend YODA-compliant layouts for derivative datasets

Workflow Platforms: Infrastructure projects implementing YODA at scale:

- **BABS**[8]: Platform implementing the "FAIRly big workflow" approach, demonstrating YODA principles for large-scale neuroimaging analyses with containerized pipelines and modular dataset composition
- **CRCNS.org**: Neuroscience data sharing platform providing YODA-structured templates and validation tools

These adoptions demonstrate that STAMPED properties solve practical organizational challenges across scales, from individual studies to community-wide infrastructure.

3.11 Convergent evolution of data-driven scientific practices

The challenges that motivate STAMPED are not unique to any single domain. Over three decades, communities spanning geophysics, statistics, genomics, neuroimaging, and machine learning independently converged on strikingly similar organizational responses to the same recurring problems: fragmented provenance, irreproducible computations, and ad-hoc project layouts. What follows is not a comprehensive review but a tracing of that convergence, organized chronologically, to show that the properties STAMPED formalizes reflect durable responses to recurring challenges.

The earliest calls for reproducibility-as-organization appeared in the 1990s. Claerbout and Karrenbach[9] demonstrated that coupling electronic documents with Makefiles could make geophysical analyses self-contained, tracked, and re-executable—anticipating Self-containment (S), Tracking (T), and Actionability (A). Buckheit and Donoho[10] extended this idea with WaveLab, a compendium packaging code, data, and figures so that "an article ... is not the scholarship itself, ... the actual scholarship is the complete software development environment and the complete set of instructions which generated the figures"—touching Self-containment (S), Actionability (A), Portability (P), and Distributability (D). Gentleman and Temple Lang[11] formalized the research compendium concept, emphasizing modular packaging for reuse and redistribution—Self-containment (S), Modularity (M), Actionability (A), and Distributability (D). Noble[12] provided the first widely adopted project-organization guide for computational biology, offering concrete directory-structure conventions that addressed Self-containment (S) and Modularity (M).

A wave of normative frameworks followed in the 2010s. Stodden[13] articulated legal and social dimensions of sharing computational research, raising concerns central to Distributability (D) and Tracking (T). Sandve et al.[14] distilled ten rules for reproducible computation—track versions, record workflows, archive environments—spanning Tracking (T), Actionability (A), and Self-containment (S). The FAIR principles[1] established findability, accessibility, interoperability, and reusability as community norms for digital objects, later extended to research software by FAIR4RS[2] and to workflows by the Workflow Community Initiative[3]. Wilson et al.[15] codified "Good Enough Practices" that operationalized many of these ideals for working scientists—Actionability (A), Tracking (T), and Self-containment (S). The YODA principles[4] combined

these insights into organizational practices from which STAMPED directly descends. Notably, the WCI-FW framework[3] extended FAIR to computational workflows but explicitly declined to define new principles, leaving the operational gap—how a research object should be structured and managed—that STAMPED fills.

Independent of these normative efforts, a parallel "Git for data" convergence occurred in tooling. git-annex (2010), Git LFS (2015), DVC (2017), Pachyderm (2014), and Icechunk (2024) all independently arrived at content-addressed identification of large files, logical separation of metadata from bulk storage, and derivation records linking inputs to outputs. Despite differing architectures—some cloud-native, some local-first, some centralized, some federated—each converged on the same core insight: that version-control semantics proven for code could and should extend to data. This convergence is evidence not of influence but of shared need, and maps most directly onto Tracking (T).

A complementary convergence emerged in platforms and metadata standards. Though not an exhaustive list: Galaxy[16] (first released 2005), Code Ocean (2017), and brainlife.io[17] each provided turnkey reproducibility services, each implementing separation of code, data, and environment as a first-class abstraction—reflecting Self-containment (S), Actionability (A), and Portability (P). On the standards side, the W3C PROV data model (2013), Frictionless Data (2007), RO-Crate[18], and BioComputeObject (IEEE 2791-2020) each addressed interoperable packaging and provenance recording. The facts that FAIR required elaboration to accommodate software[2] and workflows[3;19], and that independent standards converged on provenance and packaging, confirms that these dimensions are recurrently demanded by research practice.

Framework tooling and domain standards reveal the same pattern at the project-layout level. Cookiecutter Data Science (2016), BIDS[7] (2016), Kedro (2019), nipoppy (2023), and others, all imposed opinionated directory structures and modular separation—independently motivated by the costs of ad-hoc layouts—reflecting Modularity (M) and Self-containment (S). Each framework constrains where code and documentation reside, how raw data inputs and derived outputs are handled, which makes such projects navigable by *convention*.

Viewed together, these independent lines of development map onto every STAMPED property. Self-containment (S) appears wherever projects bundle their dependencies into a single rooted structure; Tracking (T) wherever content-addressed storage or version control records the history of changes or acquisition of every component; Actionability (A) wherever Makefiles, workflow managers, or platform APIs make re-execution or acquisition a single command; Modularity (M) wherever directory conventions or subproject composition separate concerns; Portability (P) wherever good coding practice, containers or environment specifications decouple a research object from a particular machine; Ephemerality (E) wherever computational compositions bringing together compute and data are treated as disposable and regenerable; and Distributability (D) wherever licensing, packaging standards, or federated storage enable sharing.

The formalization presented here provides a shared vocabulary for what these practices have in common, enabling researchers to evaluate and incrementally improve their workflows regardless of the specific tools they employ. Figure 1 summarizes these parallel timelines and their mapping to STAMPED properties.

3.12 Future Directions

- Yarik will jot down their ideas for such a table about AI relation - Other items such as the examples as a living entity can be briefly mentioned but do not need sub sub sections

4 Data Availability

5 Code Availability

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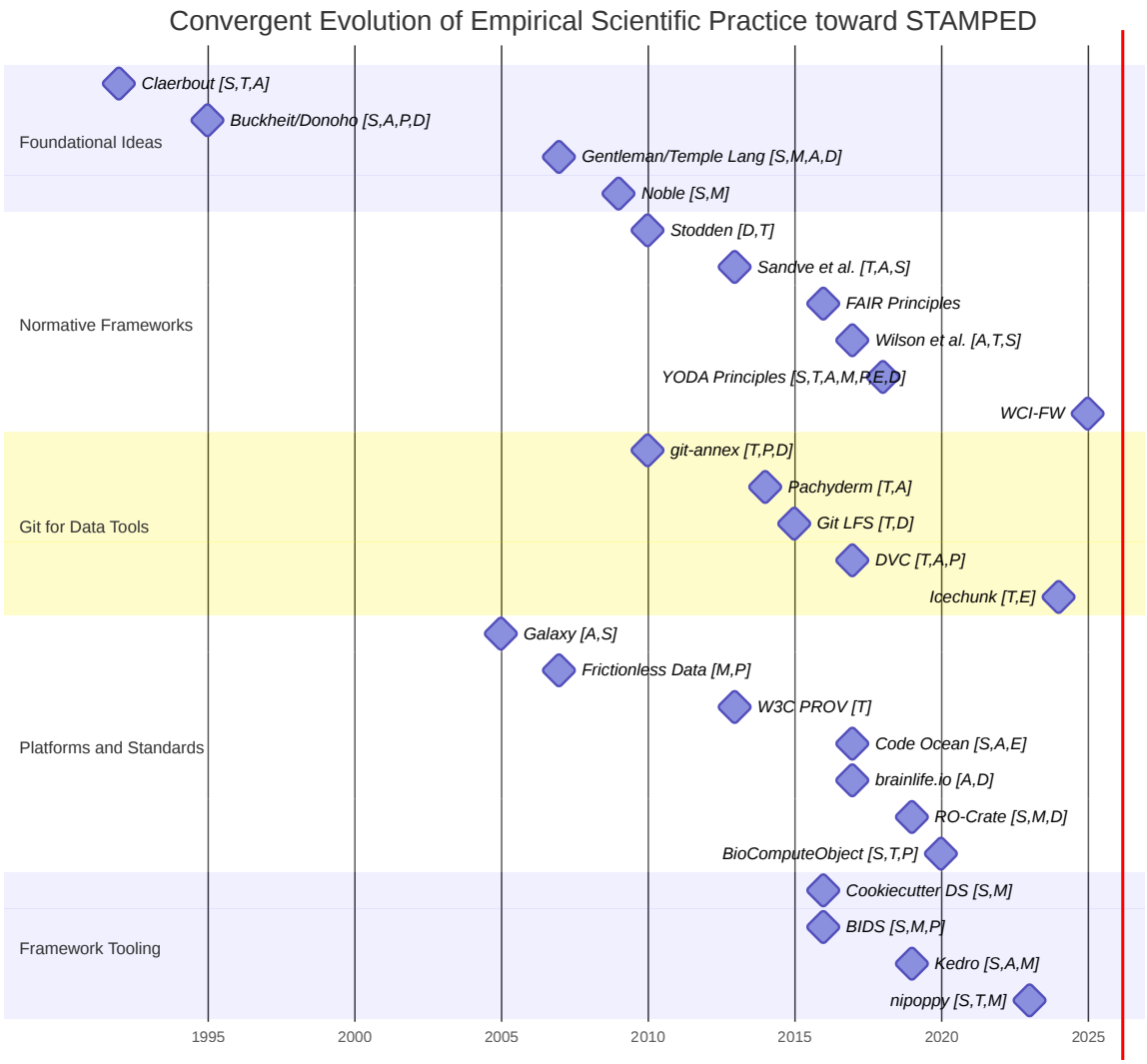

Figure 1: Convergent evolution of empirical scientific practice toward STAMPED. Five independent lines of development—foundational ideas, normative frameworks, data version-control tools, platforms and metadata standards, and framework tooling—are shown on a shared timeline from 1992 to 2025. Each milestone is annotated with the STAMPED properties it anticipates. Despite independent origins, all five tracks converged on the same organizational patterns that STAMPED formalizes.

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6 Author Contributions

7 Competing Interests

The authors declare no competing interests.

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